

● MEDIUM

● ADVANCED MATH

Equivalent expressions

10 questions · ~15 min

02

Q1

ADVANCED MATH

$$(2x+5)^2 - (x-2) + 2(x+3)$$

Which of the following is equivalent to the expression above?

A. $4x^2 + 21x + 33$

B. $4x^2 + 21x + 29$

C. $4x^2 + x + 29$

D. $4x^2 + x + 33$

Q2

ADVANCED MATH

Which expression is equivalent to $a^{\frac{11}{12}}$, where $a > 0$?

A. $\sqrt[12]{a^{132}}$

B. $\sqrt[144]{a^{132}}$

C. $\sqrt[121]{a^{132}}$

D. $\sqrt[11]{a^{132}}$

Q3

ADVANCED MATH

If $x \neq 0$, which of the following expressions is equivalent to $\frac{\sqrt{16x^4y^8}}{x^3}$?

- A. $8x^2y^4$
- B. $4xy^4$
- C. $4x^{-2}y^2$
- D. $4x^{-1}y^4$

Q4

ADVANCED MATH

Which expression is equivalent to $x^2 + 11^2 + x - 5x + 5$?

- A. $x^4 + 23x^2 - 14$
- B. $x^4 + 23x^2 + 96$
- C. $x^4 + 12x^2 + 121$
- D. $x^4 + x^2 + 146$

Q5

ADVANCED MATH

Which expression is equivalent to $\sqrt[7]{x^9y^9}$, where x and y are positive?

- A. $xy^{\frac{7}{9}}$
- B. $xy^{\frac{9}{7}}$
- C. xy^{16}
- D. xy^{63}

Q6

ADVANCED MATH

Which expression is equivalent to $\frac{8xx - 7 - 3x - 7}{2x - 14}$, where $x > 7$?

- A. $\frac{x - 7}{5}$
- B. $\frac{8x - 3}{2}$
- C. $\frac{8x^2 - 3x - 14}{2x - 14}$
- D. $\frac{8x^2 - 3x - 77}{2x - 14}$

Q7

ADVANCED MATH

Which expression is equivalent to $7x^3 + 7x - 6x^3 - 3x$?

- A. $x^3 + 10x$
- B. $-13x^3 + 10x$
- C. $-13x^3 + 4x$
- D. $x^3 + 4x$

Q8

ADVANCED MATH

Which expression is equivalent to $d - 68d^2 - 3$?

- A. $8d^3 - 14d^2 - 3d + 18$
- B. $8d^3 - 17d^2 + 48$
- C. $8d^3 - 48d^2 - 3d + 18$
- D. $8d^3 - 51d^2 + 48$

Q9

ADVANCED MATH

The expression $\frac{24}{6x+42}$ is equivalent to $\frac{4}{x+b}$, where b is a constant and $x > 0$. What is the value of b ?

- A. 7
- B. 10
- C. 24
- D. 252

Q10

ADVANCED MATH

$$(2x+3)-(x-7)$$

Which of the following is equivalent to the given expression?

- A. $x-4$
- B. $3x-4$
- C. $x+10$
- D. $2x^2+21$